Bat Algorithm-Based Energy Beamforming for Wireless Power Transfer With Dynamic Metasurface Antennas

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***Abstract*—This paper investigates the problem of energy beamforming for wireless power transfer (WPT) using dynamic metasurface antennas (DMAs). We propose a novel solution based on the bat algorithm (BA) to efficiently optimize the beamforming process. The proposed BA-based scheme enables simultaneous charging of multiple devices while avoiding power transmission in specific directions, such as areas where people or animals may be present. Our approach provides a robust and computationally efficient solution, considering key system constraints, including power transfer efficiency, antenna configurations, and DMA characteristics. Simulation results demonstrate that the BA-based method outperforms existing techniques in the literature, particularly those relying on alternating optimization, by achieving lower total power consumption and reduced computational complexity. These findings highlight the potential of the proposed method as a promising solution for future WPT systems employing DMAs.**

***Index Terms*—Energy harvesting, dynamic metasurface antenna, bat algorithm, wireless power transfer.**

I. Introduction

WIRELESS power transfer (WPT) has been of great interest in recent years due to its potential to eliminate the need for wires to power electronic devices. WPT utilizes electromagnetic fields to transfer energy from a source to a receiver without the need for physical connections, enabling a range of applications from recharging mobile devices to powering moving systems such as electric vehicles. Moreover, as the Internet of Things (IoT) continues to expand, radio frequency wireless power transfer (RF-WPT) plays a critical role in enabling the continuous operation of connected devices. Unlike other forms of WPT, such as inductive or resonant coupling, RF-WPT enables remote powering of sensors, smart appliances, and other IoT devices without the need for physical alignment or close proximity [1]. By providing a reliable power supply in these scenarios, RF-WPT, hereinafter referred to as WPT for brevity, enhances the functionality and reliability of the devices, thereby promoting a more interconnected and efficient ecosystems [2]. The integration of WPT with IoT not only improves user convenience but also supports advancements in automation and smart technology applications, paving the way for smarter cities and enhanced industrial processes [3]. As a consequence, the development of technologies to improve WPT efficiency and delivery precision has attracted lots of research [1], [2], [3].

Recently, with the advance of 5G deployment and the development of 6G technologies, there has been an increase in the operating frequency of wireless systems, which directly affects the boundary between the near-field and far-field regions. The near-field is actually subdivided into two regions: *i.)* the reactive near-field, which is very close to the antenna and where energy is associated with capacitive and inductive effects; and *ii.)* radiative near-field, where energy is propagated in spherical waveforms. While the former is not often compatible with IoT scenarios, since distances with respect to the antenna must be very short, radiative near-field WPT (NF-WPT) has emerged as a promising area of research, with great potential for delivering energy more efficiently over typical IoT distances.

One of the key elements in NF-WPT is the use of energy beamforming techniques, concentrating the energy in the location of one or multiple devices to be charged. The energy beamforming gains depend on the transmitter architecture. Fully digital structures provide the highest flexibility; nevertheless, they incur high complexity, cost, and power consumption due to the large number of required RF chains. On the other hand, analog beamforming architectures consume considerably less power, reducing the number of required RF chains. However, they suffer from poor flexibility. Therefore, hybrid architectures have emerged to provide a trade-off between complexity and flexibility of the energy beamforming [4]. In this context, waveguide-fed dynamic metasurfaces antennas (DMAs) are a promising hybrid technology [5].

The DMA is a group of configurable metamaterial elements placed on a set of waveguides, fed by RF chains, offering

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nearly-passive beamforming gains by leveraging their structural properties to optimize the distribution of electromagnetic waves [5]. These technological advancements allow more efficient energy delivery, with remarkable granularity and efficiency, especially in applications where WPT must adapt in real-time to user movement and distribution [6]. Furthermore, the metamaterial elements of the DMA are spaced more densely than half a wavelength [7], which, on the one hand, enables the modeling and control of evanescent waves near the surface, allowing for manipulations of electromagnetic waves with high energy efficiency. On the other hand, the reduced inter-element spacing also increases the mutual coupling, yielding performance losses. Therefore, unlike other reflective surfaces, the phase shift and gain provided by each element of the DMA are correlated, implying additional restrictions to the energy beamforming design.

Recently, the DMA has been investigated from a WPT perspective. For instance, in [8], an optimization framework for maximizing the energy efficiency in DMA-assisted massive MIMO uplink communications is introduced. Therein, a low-complexity algorithm is developed to enhance transmit precoding design using the DMA. In [9], the DMA architecture is leveraged as a means to facilitate the integration of a large number of metamaterial elements. Moreover, the study emphasizes its advantages in achieving simple, energy-efficient, and compact design configurations, independent of the operating frequency. The authors in [10] jointly optimize the DMA configuration and digital precoders to enhance the weighted sum harvested energy in NF-WPT, leading to improved energy efficiency with minimal emission footprint. Furthermore, in [11], a NF-WPT is analyzed, incorporating DMA and nonlinear energy harvesters. An alternating optimization approach for waveform and beam focusing is developed to minimize energy consumption while ensuring energy harvesting requirements are met. The findings indicate that increasing the number of transmission tones reduces energy consumption and that DMA outperforms the fully digital architecture in terms of energy efficiency.

A recent energy beamforming approach for WPT systems using DMAs is presented in [12], where the authors show that achieving optimal energy efficiency is challenging due to a mathematically complicated optimization problem yielded by the DMA. The authors propose an alternating optimization method that iteratively approximates the optimal solution. First, optimal digital precoders are obtained given a fixed matrix of DMA configurable weights. Then, the configurable weights are optimized, given a fixed digital precoding obtained in the previous step. The study demonstrates that alternating optimization can be used for effectively meeting the energy requirements, while the DMA outperforms the full-digital architecture in terms of minimizing the transmit power. Additionally, it is shown that transmission power decreases as the antenna array size increases but rises with operating frequency when using DMA. Nevertheless, alternate optimization incurs high computational complexity for this setup, specially as the number of variables increases. To cope with this, heuristic algorithms have emerged as alternative approaches with lower complexity [13]. Unlike reinforcement learning-based resource allocation methods that jointly optimize beamforming and passive elements in IRS-assisted Simultaneous Wireless Information and Power Transfer (SWIPT) systems [14], this work employs the BA metaheuristic to efficiently address the non-convex energy beamforming problem in DMAs operating under NF-WPT conditions. Moreover, instead of focusing on robust CSI designs as in [15], our approach emphasizes operational safety by integrating a user-avoidance constraint—essential for NF applications yet seldom explored in far-field scenarios.

Although the above-mentioned works inherently reduce the RF emission footprint by focusing the beams on the devices, they do not take into account situations where certain areas must be avoided. These scenarios are relevant when people or animals are in the same environment as IoT devices. In these cases, beams must be focused on the IoT devices while also ensuring specific areas are safe in terms of electromagnetic radiation for humans and animals [16]. For the DMA, in particular, such an issue becomes relevant once phase shifts and gains are correlated, so that the energy pollution in NF may have a negative effect in terms of human/animal exposure, or even in terms of communication interference [17].

In this paper, we address a scenario in which some devices must be charged by the DMA, while other devices or regions must be avoided by the energy beamforming, *i.e.*, they should receive power below a predefined threshold. The motivation is to avoid specific areas due to the presence of people or points of interest that need to be protected or bypassed, ensuring safe operation of RF-WPT technology. It is noteworthy that, most practical RF-WPT systems require high-power transmission to deliver meaningful energy to the intended devices, due to channel fading and propagation losses. Thus, ensuring that beam radiation stays within safe levels at some specific areas is a challenging task and also a critical prerequisite for the commercial viability of RF-WPT technology [16].

The proposed optimization considers both charging and avoidance constraints, as well as the physical characteristics of the DMA, which makes the problem inherently non-convex. To efficiently tackle this complexity, we propose a Bat Algorithm (BA)-based energy beamforming approach for NF-WPT using DMA technology. The BA is particularly suitable for global optimization tasks due to its ability to explore and exploit the search space effectively [18]. Moreover, the BA stands out among heuristic algorithms as the only one incorporating a self-adjusting zoom mechanism during the optimization process, which significantly improves convergence speed and stability while reducing the influence of algorithm parameters. Our numerical results demonstrate that the proposed BA-based scheme effectively solves the non-convex optimization problem in DMA-based NF-WPT systems, achieving fast convergence and well-focused energy distribution patterns. Compared to [12], which employs an alternating optimization method to deal with the non-convexity of the problem, the proposed BA method reduces total energy consumption at the DMA while also lowering the computational complexity of the beamforming design. Additionally, let us remark that the scenario in which certain users must be avoided is not captured by [12]. In this case, the BA successfully meets the

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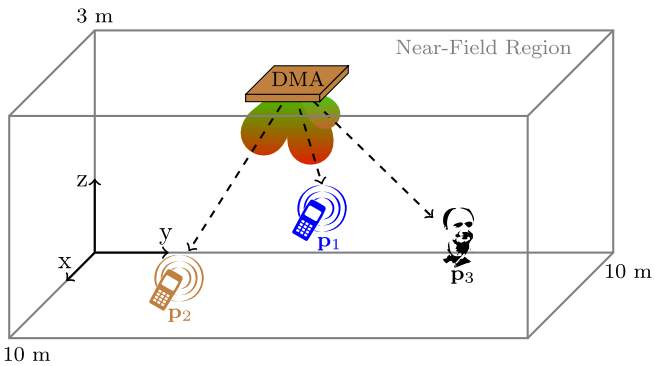

Fig. 1. WPT system with a DMA mounted on the center of the ceiling of a room, transmitting energy to $J = 2$ devices located at positions $\mathbf{p}_1$ and $\mathbf{p}_2$, while avoiding transmitting energy to $U = 1$ person located at $\mathbf{p}_3$.

system requirements, and our results reveal that overall power consumption is inversely proportional to the user avoidance threshold. Specifically, enforcing lower thresholds for the users that must be avoided increases power consumption at the DMA, whereas relaxing these constraints allows for significant power savings.

The remainder of this work is structured as follows. Section II introduces the system model employed for optimization, with the proposed BA formulation detailed in Section III. Section IV describes the considered scenarios to assess the BA's capability in solving the proposed optimization problem, presenting numerical results and discussions. Finally, the conclusions are presented in Section V.

II. System Model

We model a WPT system composed of a DMA fixed on the ceiling of a room, which serves as the energy transmitter, with a set of K single-antenna devices distributed in the three-dimensional space, as illustrated by Fig. 1. We denote the set of devices by $\mathcal{K}$, and the location of the k-th device is defined by the vector $\mathbf{p}_k = [x_k, y_k, z_k]$, where $k = 1, 2, \ldots, K$.

Furthermore, we assume that $\mathcal{K}$ is composed of two orthogonal subsets $\mathcal{J}$ and $\mathcal{U}$, so that $\mathcal{K} = \mathcal{J} \bigcup \mathcal{U}$ with $\mathcal{J} \bigcap \mathcal{U} = \emptyset$, in which $\mathcal{J}$ denotes the subset of the energy harvesting devices, while $\mathcal{U}$ is the subset of unintended devices to be avoided by the DMA.[1] Thus, defining $J = |\mathcal{J}|$ and $U = |\mathcal{U}|$ as the respective cardinality of the subsets, we have that $K = J + U$.

A. The DMA Model

The DMA model employed in this work is shown in Fig. 2, which is composed of N_v RF chains, each connected to a waveguide with N_h metamaterial elements. The number of vertical and horizontal elements can be determined by the physical antenna length L and the wavelength λ. Specifically, $N_v = \lfloor L/(\lambda/2) \rfloor$ and $N_h = \lfloor L/(\lambda/5) \rfloor$ [21]. Thus, the total number of elements of the DMA is $N_{\mathrm{D}} = N_v \times N_h$.

[1]Assuming that the unintended device has a transceiver, *e.g.*, a human carrying a mobile phone, the WPT system can estimate its position using angle-of-arrival or RSSI-based techniques [19]. In the case of an unintended device without a transceiver, alternatives such as mmWave radar sensing could be employed [20]. However, we assume that the position of the users is known to the DMA, and the method to identify or detect them is not focus of this work.

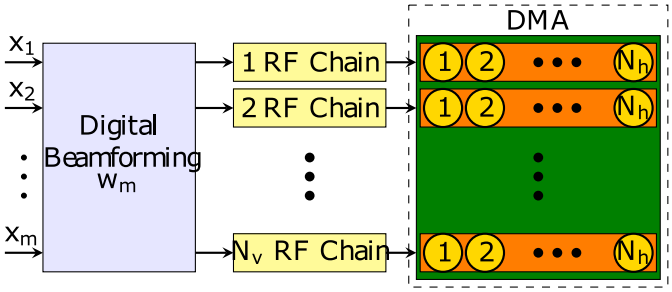

Fig. 2. Transmitter architecture based on DMA.

The signal propagation effects within the waveguide structure are described by the diagonal propagation matrix $\mathbf{H} \in \mathbb{C}^{N_{\mathrm{D}} \times N_{\mathrm{D}}}$ of the DMA, with $h_{i,l}$ being the $((i-1)N_h + l)$-th column and $((i-1)N_h + l)$-th row element of $\mathbf{H}$, where $i = 1, \ldots, N_v$ and $l = 1, \ldots, N_h$ [5], [6], [22]. In addition, the propagation coefficient for the l-th element in the i-th waveguide can be written as [8], [9], [21], [23], and [24]

$$h_{i,l} = \exp\big[-(l-1)d(\alpha_i + \mathrm{j}\beta_i)\big], \tag{1}$$

where d denotes the inter-element distance on the waveguide, α_i is the waveguide attenuation coefficient, and β_i is the propagation constant of the i-th waveguide, where $i = 1, \ldots, N_v$ and $l = 1, \ldots, N_h$. In addition, the propagation coefficients $h_{i,l}$ are frequency-dependent, as both the attenuation coefficient α_i and the propagation constant β_i vary with the operating frequency [8], [9], [21], [23], [24], [25].

Let x_m be the m-th energy symbol at the input of the digital beamforming, which is fed to the waveguides, where $m = 1, \ldots, M$, and $M = \min(N_v, J)$. Moreover, $\mathbf{w}_m \in \mathbb{C}^{N_v \times 1}$ denotes the precoding vector for x_m. Thus, considering that the energy symbols are normalized and independent, so that $\mathbb{E}\{x_n^H x_r\} = 0, \forall n \neq r$ and $\mathbb{E}\{x_n^H x_n\} = 1$, the transmit signal of the DMA is given by

$$\mathbf{s} = \sum_{m=1}^{M} \mathbf{H}\mathbf{Q}\mathbf{w}_m x_m, \tag{2}$$

where $\mathbf{Q} \in \mathbb{C}^{N_{\mathrm{D}} \times N_v}$ contains the configurable weights of the metamaterial elements, given by [21]

$$\mathbf{Q}_{(i-1)N_h + l, n} = \begin{cases} q_{i,l}, & i = n, \\ 0, & i \neq n, \end{cases} \tag{3}$$

with

$$q_{i,l} \in \mathcal{L} = \Big\{ \left(\mathrm{j} + e^{\mathrm{j}\phi_{i,l}}\right)/2 \;\Big|\; \phi_{i,l} \in [0, 2\pi] \Big\}. \tag{4}$$

This is the Lorentzian-constrained phase model, which captures the dependency between the elements' amplitude and phase. Note that $\mathcal{L}$ denotes the Lorentzian interval in the complex plane [12].

As a consequence, the transmit power of the DMA is $P^{\mathrm{TX}} = \mathbb{E}\{\mathbf{s}^H \mathbf{s}\}$, which can be written as

$$P^{\mathrm{TX}} = \sum_{m=1}^{M} \left|\mathbf{H}\mathbf{Q}\mathbf{w}_m\right|^2. \tag{5}$$

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B. Channel Model

To ensure NF propagation conditions, a device k, located at position $\mathbf{p}_k$, must be at a distance from the DMA greater than $d_{\text{fs}} = \sqrt[3]{\frac{D^4}{8\lambda}}$, which defines the Fresnel distance, and smaller than $d_{\text{fr}} = \frac{2D^2}{\lambda}$, which defines the Fraunhofer distance, where $D = \sqrt{2}L$ is the effective antenna diameter [25]. Devices positioned within this range will experience radiative NF effects, where the curvature of the electromagnetic wavefront and the non-uniform phase distribution play a significant role in shaping the signal propagation characteristics. Under these NF propagation conditions, the channel is strongly governed by deterministic line-of-sight propagation, rather than by multipath effects. This is in contrast to far-field communication scenarios, where stochastic multipath fading models are typically required to account for reflections, scattering, and diffraction.

According to [21], the channel coefficient between the l-th metamaterial element in the i-th waveguide and user k is

$$\mathscr{C}_{i,l,k} = \mathscr{A}_{i,l,k} \exp\left(-\mathrm{j}\frac{2\pi d_{i,l,k}}{\lambda}\right), \tag{6}$$

where the exponential component represents the phase variation as a function of distance, and $d_{i,l,k} = ||\mathbf{p}_k - \mathbf{p}_{i,l}||$ represents the Euclidean distance between $\mathbf{p}_k$ and the location of the l-th element in the i-th waveguide, given by $\mathbf{p}_{i,l} = [x_{i,l}, y_{i,l}, z_{i,l}]$, where $i = 1, 2, \ldots, N_v$ and $l = 1, 2, \ldots, N_h$. Also, $\mathscr{A}_{i,l,k}$ controls the amplitude of the channel coefficient, given by [26]

$$\mathscr{A}_{i,l,k} = \sqrt{F(\theta_{i,l,k}, \phi_{i,l,k})}\,\frac{\lambda}{4\pi d_{i,l,k}}, \tag{7}$$

where $(\theta_{i,l,k}, \phi_{i,l,k})$ is the elevation-azimuth pair from the i, l-th element to the k-th user, while $F(\cdot)$ is the radiation profile of each element, modeled as

$$F(\theta_{i,l,k}, \phi_{i,l,k}) = \begin{cases} 2(b+1)\cos^b(\theta_{i,l,k}), & \theta_{i,l,k} \in [0, \pi/2], \\ 0, & \text{otherwise}, \end{cases} \tag{8}$$

with b being the boresight gain.

Then, the energy signal received by the k-th device depends on the channel coefficient vector between device k and the N elements of the DMA. We denote such channel coefficient vector by $\mathbf{c}_k = [\mathscr{C}_{1,1,k}, \mathscr{C}_{1,2,k}, \ldots, \mathscr{C}_{N_v,N_h,k}]^T$, encapsulating all $l \in [1, N_h]$ elements of the $i \in [1, N_v]$ waveguides, with respect to the k-th device. Therefore, the energy signal received by the k-th device is $y_k = \mathbf{c}_k^H \mathbf{s}$, yielding a received RF power of

$$P_k^{\text{RX}} = \sum_{m=1}^{M} \left|\mathbf{c}_k^H \mathbf{H}\mathbf{Q}\mathbf{w}_m\right|^2. \tag{9}$$

Let us remark that the received RF is converted into DC power by the energy harvesting device, which is usually modeled either by non-linear or piecewise linear functions [27], [28]. Therefore, given a requirement in terms of P_k^{DC}, which is determined by the practical application [29], we need to establish the required value for P_k^{RX}, by inverting the RF-DC conversion function, to ensure maximal efficiency of the energy conversion of the rectenna at the receiver.

C. Problem Formulation

The optimization problem focuses on minimizing the transmit power P^{TX} of the DMA, while ensuring that devices $j \in \mathcal{J}$ receive power $P_j^{\text{RX}} = f^{-1}(P_j^{\text{DC}}) \geq \delta_{\text{EH}}$, at the same time that devices $u \in \mathcal{U}$ receive power $P_u^{\text{RX}} = f^{-1}(P_u^{\text{DC}}) \leq \delta_{\text{U}}$, *i.e.*, below a given threshold for safe exposure, where $f^{-1}(\cdot)$ denotes the inverse function of the non-linear energy harvesting model. This can be mathematically expressed as

$$\underset{\mathbf{Q},\mathbf{w}_m}{\text{minimize}} \quad \sum_{m=1}^{M} |\mathbf{H}\mathbf{Q}\mathbf{w}_m|^2 \tag{10a}$$

$$\text{s.t.} \quad \sum_{m=1}^{M} |\mathbf{c}_j^H \mathbf{H}\mathbf{Q}\mathbf{w}_m|^2 \geq \delta_{\text{EH}}, \quad \forall j \in \mathcal{J}, \tag{10b}$$

$$\sum_{m=1}^{M} |\mathbf{c}_u^H \mathbf{H}\mathbf{Q}\mathbf{w}_m|^2 \leq \delta_{\text{U}}, \quad \forall u \in \mathcal{U}, \tag{10c}$$

$$\mathbf{Q} \in \mathcal{L}. \tag{10d}$$

Finding the matrices $\mathbf{Q}$ and $\mathbf{w}_m$ is challenging since (10b) and (10c) are non-convex constraints. In addition, constraint (10d) imposes that the phase shifts and amplitude gains of the DMA are correlated, further complicating achieving the optimal solution.[2] To solve this optimization problem, various methods can be employed, such as convex programming, the method of Lagrange multipliers, the gradient algorithm, or alternate optimization [18]. For instance, the optimization problem in [12] is similar to the above, without constraint (10c), so that their approach alternates to find an optimal $\mathbf{w}_m$ given a fixed $\mathbf{Q}$, and then finding a suboptimal $\mathbf{Q}$ with fixed $\mathbf{w}_m$. As the authors show, the first step is a semi-definite programming (SDP) that can be solved by standard convex optimization tools such as CVX. Nevertheless, suboptimality in the second step comes from the Lorentzian and rank constraints, which must be relaxed in order to transform the problem into an SDP. As a result, the relaxed solution of the second step must be mapped to the nearest point on the Lorentzian constrained circle to become a feasible solution. By conducting this procedure iteratively, a suboptimal solution is obtained. The solutions with this method adapt well to the fact that the gain introduced by the metamaterial elements is limited and correlated with their phase values, while the gain in digital precoders can be chosen freely without any limitation and correlation with the phase values, making them the dominant factor in determining the transmit power.

However, let us remark that the initialization is a typical issue for alternate algorithms like [12], for which finding first a feasible solution is usually handled by metaheuristic algorithms. Alternatively, given the fact that relaxation techniques are needed to deal with the Lorentzian constraints, in the following we adopt a heuristic approach based on the BA to solve the optimization problem in (10). Among heuristic algorithms, it stands out due to its distinctive self-adjusting zoom capability. This feature allows the BA to dynamically narrow the search space around the best solution found so

[2]Let us also remark that $\mathbf{H}$ is assumed to be known in the current work, similar to [9], [10], [11], and [12], while the additional steps of estimating $\mathbf{H}$ as well as the regions to be avoided are left as future work.

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far, progressively refining candidate solutions and increasing the likelihood of converging to an optimal or near-optimal result [18].

III. Bat Algorithm

The BA, developed by Yang [13], is inspired by the echolocation behavior of bats, which emit sound pulses and analyze the returning echoes to detect prey, obstacles, and other environmental features in dark settings. BA is effective in handling complex, high-dimensional problems due to its ability to balance exploration and exploitation within the solution space. Compared to other heuristic methods, such as the genetic algorithm or Particle Swarm Optimization (PSO), in [18] the BA has demonstrated better performance in finding high-quality solutions in fewer iterations. Importantly, the BA is the only method that incorporates a self-adjusting zoom mechanism during the optimization process, which significantly improves convergence speed and stability; for this reason, it was chosen for optimizing systems with DMAs.

The summarized steps of the BA is given as follows:

1) There are N bats, whose position changes at each iteration t;
2) The position of each bat ($\mathbf{X}_n$) represents a potential solution to the optimization problem, *i.e.*, the optimized reconfigurable weights $\mathbf{Q}$ of the DMA elements and the precoding vector $\mathbf{w}_m$;
3) At each iteration, $\mathbf{X}_n$ changes according to the frequency of the pulse emitted by the bat ($\mathbf{F}_n$), which controls the size of the search space, and a velocity $\mathbf{V}_n$, which takes into account the best position of the colony up to the current iteration;
4) The position of each bat also depends on a pulse emission rate parameter (r_n), which defines if the bat adopts global or local searches. In addition, r_n is updated with t, so that the bats concentrate on local searches in the long term;
5) Each new position of the bats is validated against the optimization problem constraints in (10). New positions are accepted only if P^{TX} is reduced and the constraints $P^{\mathrm{RX}}_{j,n}[t] \geq \delta_{\mathrm{EH}}$ and $P^{\mathrm{RX}}_{u,n}[t] \leq \delta_{\mathrm{U}}$ are satisfied;
6) The solutions provided by each bat are compared at the end of each iteration, updating the overall best solution.

A. Initialization

The colony is composed of N bats and their movement is discretized over time and updated after each iteration t. Each bat in the colony is assigned an index n, whose position represents a potential solution to the optimization problem. In other words, the position $\mathbf{X}_n$ of the n-th bat at the t-th iteration concatenates the reconfigurable weights of the metasurface elements $\mathbf{Q}$ with the precoding vector of each bat, $\mathbf{w}_n$. Thus, to match the dimensions of the vectors, we first define $\mathbf{W}_n = \mathbf{w}_n\mathbf{w}_n^H \in \mathbb{C}^{N_v \times N_v}$, so that

$$\mathbf{X}_n = \begin{bmatrix} \mathbf{Q} \\ \mathbf{W}_n \end{bmatrix} \quad \in \mathbb{C}^{(N_{\mathrm{D}}+N_v)\times N_v}, \tag{11}$$

in which $\mathbf{w}_n$ can be re-obtained by taking the eigenvectors of $\mathbf{W}_n$ multiplied by the square root of their respective eigenvalues. Thus, for each bat n at the iteration t we have the transmission power given by

$$P^{\mathrm{TX}}_n[t] = |\mathbf{HQw}_n|^2, \tag{12}$$

while the power received by nodes $j \in \mathcal{J}$ and $u \in \mathcal{U}$ are respectively given by

$$P^{\mathrm{RX}}_{j,n}[t] = |\mathbf{c}_j^H\mathbf{HQw}_n|^2, \quad \forall j \in \mathcal{J}, \tag{13}$$

$$P^{\mathrm{RX}}_{u,n}[t] = |\mathbf{c}_u^H\mathbf{HQw}_n|^2, \quad \forall u \in \mathcal{U}. \tag{14}$$

In addition, it is important to highlight that the search space for $\mathbf{X}_m$ assumes the interval of the Lorentzian distribution, while the restriction $\mathbf{Q} \in \mathcal{L}$ in (10d) is further checked based on the elements of $\mathbf{X}_n$ that correspond to the optimized $\mathbf{Q}$ according to (11).

B. Movement of the Bats (Potential Optimized Solutions)

1) Frequency and Velocity Update: Each bat n of the colony is randomly initialized with a position $\mathbf{X}_n[0]$, velocity $\mathbf{V}_n[0]$, and frequency $\mathbf{F}_n[0]$. At each time step t, these variables are updated as follows. The frequency of each bat is defined within a predefined range

$$\mathbf{F}_n[t] = \mathbf{F}_{\min} + (\mathbf{F}_{\max} - \mathbf{F}_{\min}) \circ \boldsymbol{\Upsilon}, \tag{15}$$

where $\mathbf{F}_{\min}$ and $\mathbf{F}_{\max}$ are matrices representing the minimum and maximum pulse emission frequencies, respectively, $\boldsymbol{\Upsilon}$ is a matrix composed of uniformly distributed random values in $[0, 1]$, while $\circ$ is the Hadamard product. Here, all matrices match the size of the position matrix $\mathbf{X}_n$.

Based on $\mathbf{F}_n[t]$, the new velocity is calculated as

$$\mathbf{V}_n[t+1] = \mathbf{V}_n[t] + (\mathbf{X}_n[t] - \mathbf{X}^\star) \circ \mathbf{F}_n[t], \tag{16}$$

where $\mathbf{X}^\star$ is the best position of the colony up to the current iteration. Let us remark that all bats have different positions during the operation of the BA, while $\mathbf{X}^\star$ represents the best position among all N bats.

2) Global vs. Local Search: The new position of each bat, denoted as $\mathbf{X}_n[t+1]$, is computed based on the flying strategy. The flying strategy of the n-th bat is defined according to the pulse emission rate $r_n[t]$, so that the bat adopts a *global search* flight mode with probability $(1 - r_n[t])$, and a *local search* flight mode with probability $r_n[t]$.

Then, when the bat is in global search flight mode, the new position is calculated by

$$\mathbf{X}_n[t+1] = \mathbf{X}_n[t] + \mathbf{V}_n[t+1], \tag{17}$$

while in local search flight mode, the new position is

$$\mathbf{X}_n[t+1] = \mathbf{X}^\star + \boldsymbol{\Omega}\bar{A}[t], \tag{18}$$

where $\boldsymbol{\Omega}$ is a matrix with elements in the range $[-1, 1]$ generated randomly with normal distribution and $\bar{A}[t]$ is a scalar containing the average pulse amplitudes.

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3) Lorentzian Constraint: To ensure that $\mathbf{X}_n$ complies with the Lorentzian constraint, a quantization process based on sampling was implemented. First, a reference vector $\mathbf{q}$ using the Lorentzian profile in (4) is generated, containing φ equally spaced samples of $\phi_{i,l} \in [0, 2\pi]$. Then, by taking only the first N_{D} rows of $\mathbf{X}_n$ resulting from the algorithm, which correspond to the optimized $\mathbf{Q}$ according to (11), the closest value within the Lorentzian sample vector $\mathbf{q}$ is identified, *i.e.*,

$$\tilde{x}_{i,j} = \arg\min_{q \in \mathbf{q}} |x_{i,j} - q|, \tag{19}$$

where $x_{i,j} \in \mathbf{X}_n$, $i \in [1, N_{\mathrm{D}}]$, $j \in [1, N_v]$. Then, $\tilde{x}_{i,j}$ replaces $x_{i,j}$ in $\mathbf{X}_n$, ensuring that all elements are consistent with the physical behavior imposed by the Lorentzian constraint.

C. Validation and Update of the Best Solution of the Colony

Furthermore, notice that the index $t+1$ is used in (16)-(18) to denote new velocity and new positions of the bats. This is due to the fact that new velocities and positions are not always accepted by the BA. New positions are accepted only if P^{TX} is reduced and the constraints $P^{\mathrm{RX}}_{j,n}[t] \geq \delta_{\mathrm{EH}}$ and $P^{\mathrm{RX}}_{u,n}[t] \leq \delta_{\mathrm{U}}$, given in (10b)–(10c) are satisfied.

Specifically, each bat represents a candidate solution for the optimization problem, so that we calculate $P^{\mathrm{TX}}_n[t+1]$ in (12) based on $\mathbf{X}_n[t+1]$, *i.e.*, the power consumption of the DMA yielded by the solution provided by the n-th bat. If $P^{\mathrm{TX}}_n[t+1]$ is smaller than the best solution already achieved by that bat ($P^{\mathrm{TX}}_n[t+1] < P^{\star}_n$), then $\mathbf{X}_n[t+1]$ is accepted, while the pulse emission rate and the loudness are updated. Furthermore, if the solution provided by the n-th bat is the best among all bats, *i.e.*, $P^{\star}_n < P^{\star}$, where $P^{\star}$ denotes the best solution provided among all bats up to the current iteration, then the global variables $P^{\star}$ and $\mathbf{X}^{\star}$ are updated.

D. Self-Adjusting Zoom Mechanism

The pulse emission rate $r_n[t]$, which dictates the flying mode of the n-th bat, is calibrated by the parameter $\gamma \in [0, 1]$ according to

$$r_n[t+1] = r_n^{\max}\left[1 - \exp\left(-\gamma t\right)\right], \tag{20}$$

where $r_n^{\max}$ is the maximum attainable pulse emission rate. Notice that r_n increases with t, so that the bats tend to perform local searches more often in the long term. In addition, $\gamma \leq 1$ is employed to reduce the increase slope; otherwise, only local searches are performed by the algorithm. Thus, a smaller γ makes the increase slope slower, while increasing γ also makes r_n increase faster. Furthermore, the loudness is updated according to

$$A_n[t+1] = \xi A_n[t], \tag{21}$$

where $\xi < 1$ is a constant. Thus, at each iteration $A_n[t+1]$ is reduced with respect to $A_n[t]$ proportionally to ξ, narrowing the local search space.

Let us remark that a key and unique feature of BA is such self-adjusting zoom mechanism, which dynamically balances exploration and exploitation throughout the optimization process. Unlike other heuristic algorithms, BA automatically adjusts its search scale: performing a wide search over the solution space (*zoom-out*) in the early stages to explore diverse regions, and gradually focusing on local refinement (*zoom-in*) as the population converges. This process is guided by the average pulse amplitude ($\bar{A}[t]$) and the pulse emission rate ($r_n[t]$). Consequently, BA is uniquely robust, significantly less sensitive to manual parameter tuning, and ensures fast and stable convergence, outperforming other swarm-based methods in challenging optimization scenarios [18].

TABLE I
COMPLEXITY ORDER OF THE BA AND CVX SDP-BASED SOLVERS

Algorithm	Complexity Order
Bat Algorithm (BA)	$\mathcal{O}(N\mathscr{D})$
CVX SDP	$\mathcal{O}(\mathscr{D}^3 \log(1/\epsilon))$

Finally, it is also important to mention that the BA, as well as other metaheuristic optimization algorithms, do not guarantee convergence to the global optimum in a strict mathematical sense, but rather rely on randomized mechanisms to explore the search space efficiently [30]. Metaheuristics aim to achieve good enough solutions for complex optimization problems where exact methods are impractical or computationally prohibitive [31].

E. Algorithm

Algorithm 1 summarizes the proposed BA-based optimization solution for the DMA-assisted WPT system.

F. Complexity Order

The complexity order of the BA is given by the product of the number of bats in the colony (N) and the problem dimension ($\mathscr{D}$) [32]. On the other hand, CVX is a widely used tool for solving convex optimization problems. Following [33], the complexity order of CVX depends on the problem dimension and on $\log(1/\epsilon)$, where ϵ is the algorithm's precision, usually in the order of 10^{-8}. Furthermore, for SDP matrices, as it is the case in [12], the complexity order depends on the Cholesky decomposition of the matrices, with complexity order scaling with $\mathscr{D}^3$ [34].

Using the Big $\mathcal{O}$ notation, Table I summarizes the complexity orders of the two algorithms. Notice that the proposed BA depends less on the problem dimension, and it is less complex than CVX SDP as long as the number of bats is $N < \mathscr{D}^2 \log(1/\epsilon)$.

IV. RESULTS

Here, the simulations employed a frequency of 5 GHz and an antenna length of $L = 0.2$ m. The DMA was fixed on the ceiling, located at the center of a flat region measuring 10×10 meters, at a height of 3 meters at position $[5, 5, 3]$. For the BA, the initial iteration is set to $t = 0$, and the maximum number of iterations is $t_{\max} = 1000$. The initial volume of the pulses is set to $A_n[0] = 1$, and the initial pulse emission rate is $r_n[0] = 1$, $\forall n$. The algorithm includes the parameters $\xi = 0.5$, $\gamma = 0.9$, and $r_n^{\max} = 1$ which are frequently used in

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Algorithm 1 BA-Based Optimization Algorithm

Input: N, $\mathcal{J}$, $\mathcal{U}$, $\mathbf{p}_k$, δ_{EH}, δ_{U}, $\mathbf{F}_{\max}$, $\mathbf{F}_{\min}$, $r_n^{\max}$, γ, ξ, φ
Initialize:
$r_n[0] = 1$, $A_n[0] = 1$ for each bat;
Initialize the Lorentzian sample vector $\mathbf{q} \in \mathcal{L}$ using (4), containing φ equally spaced samples of $\phi_{i,l} \in [0, 2\pi]$;
Initialize population of bats for $n = 1, \ldots, N$;
Set `feasible` $\leftarrow$ `false`; ▷flag for feasible solution
Evaluate the objective function (10a) for each bat positions $\mathbf{X}_n$, ensuring constraint (10d);
Set the initial best position $\mathbf{X}^\star$ and the initial power consumption $P^\star$;
while $t \leq t_{\max}$ **do**
  for each bat $n = 1, \ldots, N$ **do**
    Calculate $\mathbf{F}_n[t]$ using (15);
    Update $\mathbf{V}_n[t+1]$ using (16);
    if $r_n[t] <$ `rand` **then**
      Calculate $\mathbf{X}_n[t+1]$ using (17);
    else
      Calculate $\mathbf{X}_n[t+1]$ using (18);
    end if
    for each element $x_{i,j} \in \mathbf{X}_n[t+1]$ **do**
      $\tilde{x}_{i,j} \leftarrow \arg\min_{q \in \mathbf{q}} |x_{i,j} - q|$;
      $x_{i,j} \leftarrow \tilde{x}_{i,j}$;
    end for
    Evaluate $P_n^{\mathrm{TX}}[t]$, $P_{j,n}^{\mathrm{RX}}[t]$, $\forall j \in \mathcal{J}$, and $P_{u,n}^{\mathrm{RX}}[t]$, $\forall u \in \mathcal{U}$ using (12)-(14);
    if $(P_n^{\mathrm{TX}}[t] < P_n^\star)$ & $(P_{j,n}^{\mathrm{RX}}[t] \geq \delta_{\mathrm{EH}})$ & $(P_{u,n}^{\mathrm{RX}}[t] \leq \delta_{\mathrm{U}})$ **then**
      Accept the new solution $\mathbf{X}_n[t+1]$;
      Update $r_n[t+1]$ and $A_n[t+1]$ using (20)-(21);
      `feasible` $\leftarrow$ `true`; ▷mark feasible solution found
      if $P_n^{\mathrm{TX}}[t] < P^\star$ **then**
        Update $\mathbf{X}^\star$ and $P^\star$;
      end if
    end if
  end for
  if `feasible = false` **then**
    break ▷stop if no feasible solution found
  end if
end while
Output: $P^\star$, $\mathbf{X}^\star$

the literature to control the search dynamics [35]. The number of bats is set to $N = 500$.

Furthermore, the values of $\mathbf{F}_{\min}$ and $\mathbf{F}_{\max}$ are configured to define the search space for amplitude, ranging from -0.5 to 0.5, and for phase, ranging from 0 to 1. However, to ensure compliance with the constraint in (10d), the values obtained within this space are further mapped to the Lorentzian circle by means of (19) constructing a vector $\mathbf{q}$ with $\varphi = 5000$ equally spaced samples of $\phi_{i,l} \in [0, 2\pi]$. This discretization corresponds to a quantization step of $2\pi/5000 \approx 0.00126$ rad, which is sufficient for the search space to closely approximate a continuous domain, making the quantization effects negligible. Thus, although the initial search space is broader than the imposed Lorentzian constraint, the algorithm effectively operates within the limits defined by (10d). This approach preserves Lorentzian dynamics, allowing the algorithm to efficiently explore joint variations in amplitude and phase, ensuring that the optimization process remains synchronized and converges towards optimal solutions [5], [11].

A. Description of the Scenarios

To evaluate the system's performance, two distinct types of scenarios are tested, as illustrated in Fig. 1, which depicts the spatial arrangement of the users within the system. Each scenario is designed to simulate different real-world conditions of WPT:

- **Charging Scenarios**: These scenarios focus solely on delivering energy to the intended users.
  - **Scenario 1**: A single device ($J = 1, U = 0$) at position $\mathbf{p}_1 = [4, 5, 1]$ is receiving wireless power, representing an isolated charging scenario.
  - **Scenario 2**: Two devices ($J = 2, U = 0$) at positions $\mathbf{p}_1 = [5, 5, 1]$ and $\mathbf{p}_2 = [6.5, 4, 1]$ are receiving wireless power simultaneously, evaluating the system's ability to handle multiple users effectively.
- **Charging and Avoidance Scenarios**: These scenarios include both delivering energy to intended users and avoiding energy transfer to unintended users.
  - **Scenario 3**: One device at position $\mathbf{p}_1 = [4, 5, 1]$ is receiving wireless energy transfer, while another at position $\mathbf{p}_2 = [5, 5, 1]$ is being avoided ($J = 1, U = 1$), simulating a partially distributed energy allocation scenario.
  - **Scenario 4**: Two devices at positions $\mathbf{p}_1 = [5, 5, 1]$, $\mathbf{p}_2 = [6.5, 4, 1]$ are receiving wireless energy transfer, while a third device at $\mathbf{p}_3 = [7, 6, 1]$ is being avoided ($J = 2, U = 1$), representing a more challenging scenario in terms of energy beamforming.

The position of the devices are initially fixed for illustration purposes, so that we can evaluate the received power in the (x, y) plane where the users are located, clearly comparing the scenarios where users must be avoided or not in Sections IV-B and IV-C, respectively. Nevertheless, an evaluation where positions are random is also carried out in Section IV-D.

B. Charging Scenarios

Fig. 3 illustrates the transmission power required at the DMA as a function of the iterations of the proposed BA approach for Scenarios 1 and 2 with the power threshold defined as $\delta_{\mathrm{EH}} = 350\ \mu$W. The choice for δ_{EH} is motivated by the rectenna nonlinear modeling in [27] using the experimental data from [36], which indicates 350 μW as the received power that yields the maximal RF-to-DC conversion efficiency for the rectenna. In addition, a PSO algorithm has also been implemented under the same conditions for comparison. As we observe, the BA requires fewer than 200 iterations to optimize

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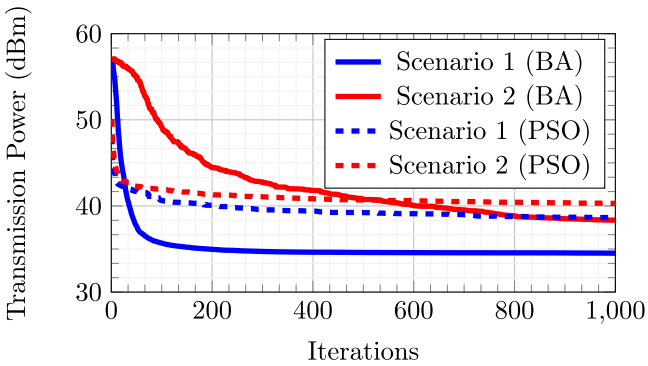

Fig. 3. DMA transmission power vs. the number of iterations of the BA for Scenarios 1 and 2, compared to the PSO algorithm.

TABLE II

TRANSMISSION POWER REQUIRED AT THE DMA OBTAINED USING THE PROPOSED BA, THE SDP APPROACH IN [12], AND THE PSO ALGORITHM, FOR AN OPERATING FREQUENCY OF 5 GHZ AND $\delta_{\text{EH}} = 350\ \mu$W

Algorithm	**Scenario 1** ($K = 1$)	**Scenario 2** ($K = 2$)
BA	34.51 dBm	38.34 dBm
SDP in [12]	43.05 dBm	44.46 dBm
PSO	38.62 dBm	40.3 dBm

Scenario 1, becoming stable beyond that value. For Scenario 1, the transmission power stabilizes at $P^{\text{TX}} = 34.51$ dBm, while $P^{\text{TX}} = 38.34$ dBm for Scenario 2. It is also important to mention that the proposed BA approach is able to maintain the threshold power of δ_{EH} in both scenarios. In Scenario 1, a lower transmission power is achieved compared to Scenario 2, since providing power to more devices implies additional energy. Furthermore, compared to PSO, BA is able to reduce the required transmission in both scenarios. It is also worth mentioning that the complexity order of PSO is given by the product of the number of particles and the problem dimension [37]. Therefore, we employ the BA and PSO with the same computational complexity by setting the number of particles in PSO equal to the number of bats (N) in BA.

These results highlight the BA as an effective solver for this convex problem, as evidenced by the comparison with the results in [12] and the PSO algorithm, summarized in Table II. Compared to [12], the proposed method reduces the required transmission power by at least 6 dB. Furthermore, when compared to PSO, the BA is able to reduce the required transmission power by 4 dB in Scenario 1, and by 2 dB in Scenario 2, as shown in Table II. Unlike convex optimization methods, which often rely on iterative solvers with higher computational overhead, the BA algorithm performs a more direct exploration of the solution space. This allows the proposed method to efficiently identify power-optimal configurations without the need for complex mathematical transformations or relaxation techniques, especially in scenarios with high-dimensional or non-convex constraints.

Additionally, Fig. 4 and Fig. 5 illustrate the influence of γ and ξ on the convergence behavior of the BA in terms of the transmit power over the iterations. We consider only Scenario 1 in these figures, although a similar behavior is observed for the other scenarios. As we observe, both γ and ξ slightly impact the convergence speed, the selection for $\gamma = 0.9$ and $\xi = 0.5$ achieving the optimized performance.

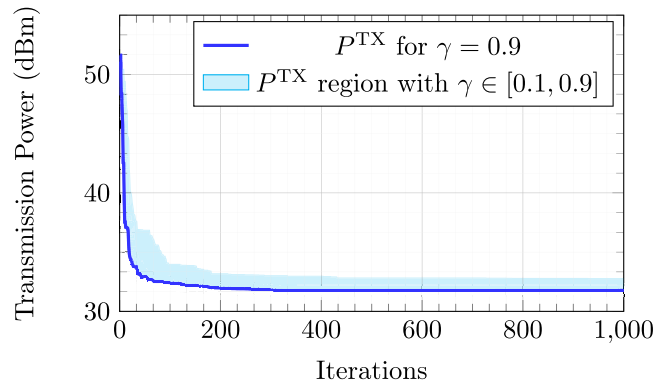

Fig. 4. Region of γ variation from 0.1 to 0.9 (shaded), and selected γ value.

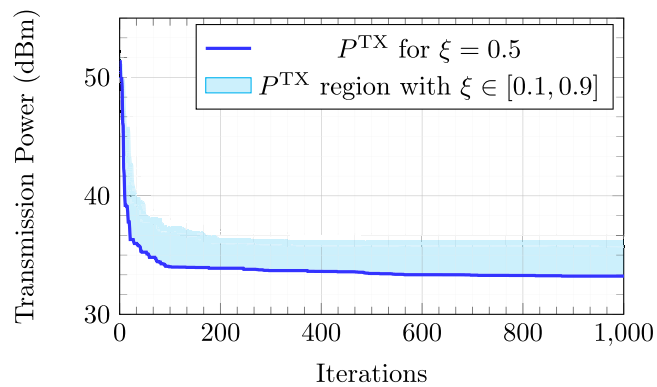

Fig. 5. Region of ξ variation from 0.1 to 0.9 (shaded), and selected ξ value.

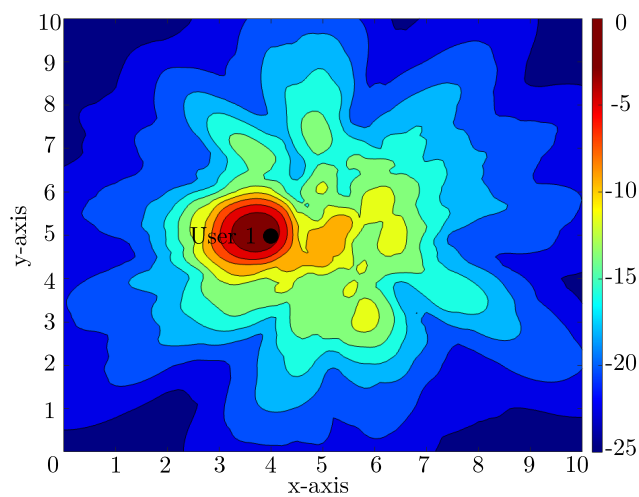

Fig. 6. Received power, in dBm, in the (x, y) plane at the devices' location ($z = 1$) for Scenario 1.

Fig. 6 illustrates the received power in the (x, y) plane where the users are located for Scenario 1, while Fig. 7 illustrates it for Scenario 2. With this approach, it is possible to visualize how the energy beamforming behaves at $z = 1$, *i.e.*,

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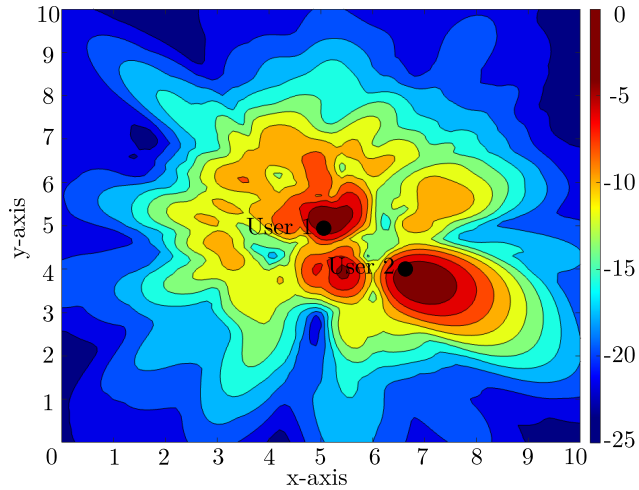

Fig. 7. Received power, in dBm, in the (x, y) plane at the devices' location $(z = 1)$ for Scenario 2.

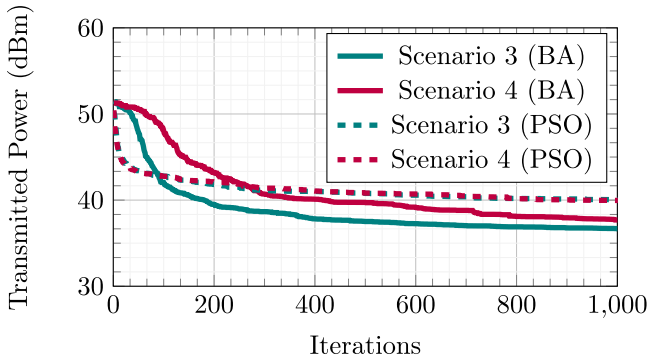

Fig. 8. DMA transmission power vs. the number of iterations of the BA for Scenarios 3 and 4, compared to the PSO algorithm.

in the plane where the users are positioned.[3] As we observe by the color gradient in both figures, the users are located in a region where the power is above $\delta_{\mathrm{EH}} = 350\ \mu\mathrm{W} = -4.56$ dBm, which is the optimization constraint of BA. Let us also highlight that Figs. 6 and 7 show that energy spreads in different positions other than that of the intended devices, which is inherent to DMAs where phase/gain optimization is constrained in a different way than in traditional arrays. Therefore, the motivation for including restriction (10c) in the optimization problem is an important characteristic for the DMA, while in traditional arrays most energy can be very well focused on the intended devices.

C. Charging and Avoidance Scenarios

Fig. 8 illustrates the transmission power required at the DMA as a function of the iterations of the proposed BA approach for Scenarios 3 and 4, in which $K = 1$ or $K = 2$ devices are being charged, respectively, while $U = 1$ device must be avoided. The power thresholds are defined as $\delta_{\mathrm{EH}} = 350\ \mu\mathrm{W}$ and $\delta_{\mathrm{U}} = 100\ \mu\mathrm{W}$. Such cases are not

[3]Let us remark that the proposed BA scheme only handles the point-wise constraints rather than the entire plane. Thus, the δ_{EH} constraints are met only at device positions, while the remainder of the (x, y) plane represents the spread of the energy beamforming.

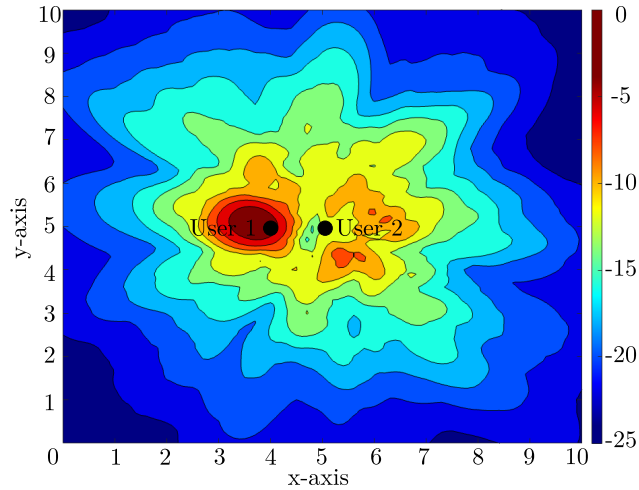

Fig. 9. Received power, in dBm, in the (x, y) plane at the devices' location $(z = 1)$ for Scenario 3.

often considered by the literature and increase the complexity of the optimization problem. This additional constraint significantly restricts the feasible solution space. Consequently, the BA must explore the solution space more carefully to avoid violating constraints, leading to a higher number of iterations. This highlights the trade-off between the number of constraints and convergence speed. While the proposed algorithm effectively handles both charging and avoidance requirements, the problem complexity increases as more users or regions need to be avoided. These aspects are critical in NF-WPT scenarios, where precise beam control is necessary for safe and efficient operation.

As we observe from the figure, in order to maintain the received power below δ_{U} for the device to be avoided, around 800 iterations are needed by the proposed algorithm to converge, in order to handle the additional constraint.[4] After convergence, the transmission power of the DMA with the proposed BA scheme stabilized at $P^{\mathrm{TX}} = 36.67$ dBm in Scenario 3 and $P^{\mathrm{TX}} = 37.68$ dBm in Scenario 4, outperforming the PSO algorithm in the same situations. Unlike other works from the literature, which do not account for these situations, the ability to impose a restriction to avoid certain individuals is a contribution of the proposed method.

Fig. 9 illustrates the received power in the (x, y) for the Scenario 3, in which the device denoted by "User 1" is being charged with a threshold $\delta_{\mathrm{EH}} = 350\ \mu\mathrm{W} = -4.56$ dBm, while "User 2" must be avoided, with a threshold $\delta_{\mathrm{U}} = 100\ \mu\mathrm{W} = -10$ dBm. As we observe by the color gradient of the figure, both users are located in a region where the optimization constraints of the BA are met. Comparing it with Fig. 6, User 1 is in the same position $\mathbf{p}_1 = [4, 5, 1]$, while User 2 has been positioned in a spot where $P_u^{\mathrm{RX}} = -7.9$ dBm in that situation. Therefore, Fig. 9 shows that the proposed BA-algorithm is able to meet the constraint $P_u^{\mathrm{RX}} \leq \delta_{\mathrm{U}}$ for User 2.

[4]In terms of implementation, we have noticed that omitting constraint (10c) for the first 40 iterations, and then re-introducing it after that, slightly increases the convergence speed for Scenarios 3 and 4.

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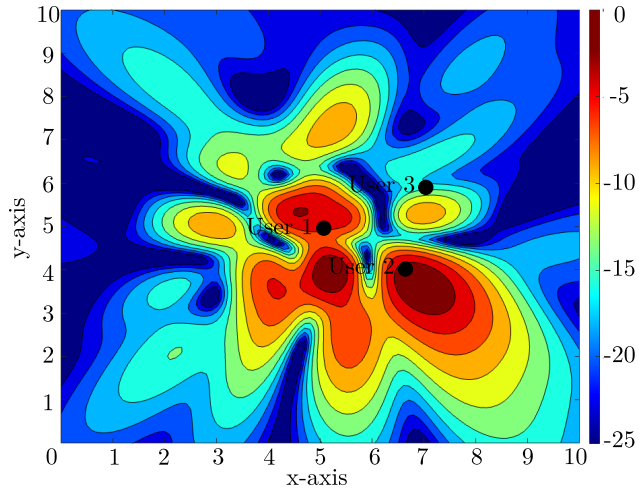

Fig. 10. Received power, in dBm, in the (x, y) plane at the devices' location $(z = 1)$ for Scenario 4.

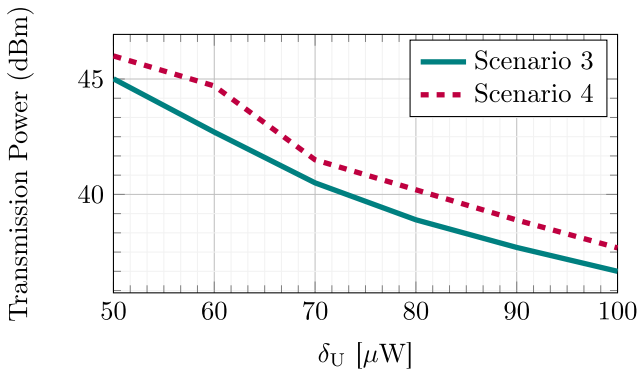

Fig. 11. DMA transmission power as a function of the threshold δ_{U} for Scenarios 3 and 4.

Similarly, Fig. 10 illustrates Scenario 4, in which "User 1" and "User 2" are being charged, while "User 3" must be avoided, with similar conclusions obtained from the color gradient. Comparing it Fig. 7, Users 1 and 2 are in the same positions $\mathbf{p}_1 = [5, 5, 1]$ and $\mathbf{p}_2 = [6.5, 4, 1]$, while User 3 has been positioned at $\mathbf{p}_3 = [7, 6, 1]$, which yielded $P_u^{\mathrm{RX}} = -9.6$ dBm in Fig. 7. As we observe, the energy spread is considerably different in Fig. 10 compared to Fig. 7. This is due to the fact that all users in this example lie in the surroundings of the boresight direction of the array (point $[5, 5, 1]$). In uniform array configurations such as DMAs, the boresight and its surroundings naturally concentrate a significant portion of the radiated energy due to the constructive contribution of the array elements in that direction. Thus, creating a null near the boresight is notoriously more challenging than in other spatial regions, since the steering vectors corresponding to directions near the main axis tend to exhibit high correlation. As a result, the beam configuration considerably changes in order to meet both charging and avoiding thresholds of the optimization problem. Therefore, Fig. 10 highlights how challenging Scenario 4 becomes under such conditions, while the proposed method is still able to meet the constraints.

Finally, Fig. 11 plots P^{TX} as a function of δ_{U} for Scenarios 3 and 4, for which we observe that the avoiding power threshold considerably impacts the transmission power. If a stricter δ_{U} is established, the transmission power of the DMA must be considerably increased. On the other hand, relaxing δ_{U} decreases P^{TX}. For instance, with $\delta_{\mathrm{U}} = 100$ μW, $P^{\mathrm{TX}} = 36.67$ dBm is required for the Scenario 3 and $P^{\mathrm{TX}} = 37.68$ dBm is required for the Scenario 4, which is very similar to that obtained in Scenarios 1 and 2. Nevertheless, the transmission power increases to $P^{\mathrm{TX}} = 45.01$ dBm and $P^{\mathrm{TX}} = 46$ dBm, respectively for Scenarios 3 and 4, when $\delta_{\mathrm{U}} = 50$ μW.

TABLE III

AVERAGE TRANSMIT POWER OF THE DMA FOR 30 RANDOM POSITIONS USING THE PROPOSED BA SCHEME, COMPARED TO PSO

	BA	PSO
Scenario 1	37.52 dBm	40.96 dBm
Scenario 2	37.39 dBm	42.09 dBm
Scenario 3	37.34 dBm	41.90 dBm
Scenario 4	37.43 dBm	41.76 dBm

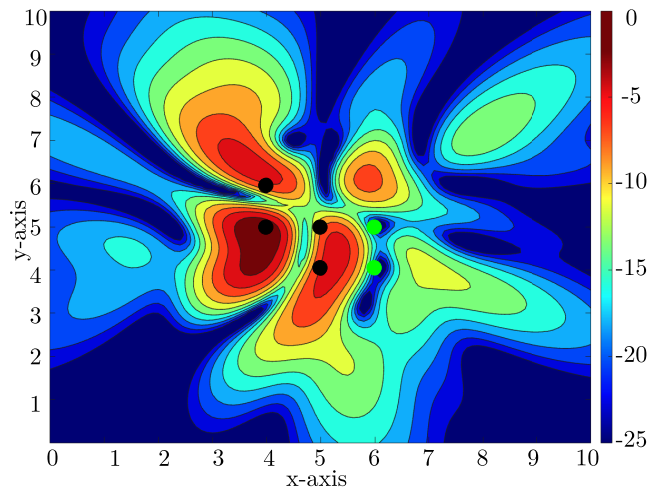

Fig. 12. Received power, in dBm, in the (x, y) plane at the devices' location $(z = 1)$ with $J = 4$ (black dots) and $U = 2$ (green dots).

D. Other Scenarios

In the previous subsections, the position of the devices are fixed for illustration purposes, so that we can clearly show that both energy delivery and avoidance can be challenging in practice. However, this is not a limitation of the proposed BA, as it can optimize beamforming for random device positions. Table III shows the average performance of the proposed BA scheme for 30 random positions employed in Scenarios 1–4. In addition, we have also added the PSO algorithm for comparison. As we observe, the average of the obtained transmit power for the DMA is quite similar as in the previous subsections. In addition, a performance improvement around 4 dBs is observed compared to the PSO algorithm, demonstrating its superiority in terms of reducing the transmitted power.

In addition, increasing the number of users directly increases the number of constraints in the optimization problem, thereby increasing its complexity. The number of user that the DMA can handle actually depends on the size of the DMA, and

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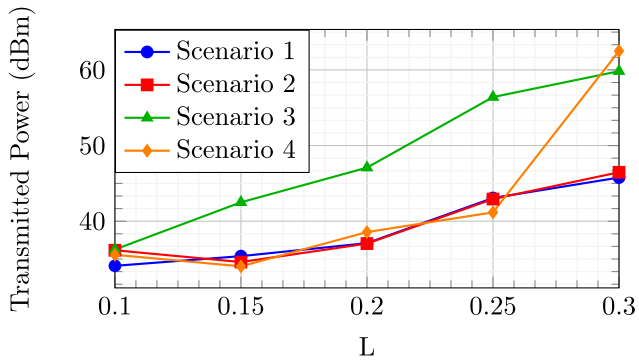

Fig. 13. Transmission power as a function of L (in meters).

increasing the number of devices would make the BA to adjust the same set of variables to satisfy all user-related constraints. While the number of variables handled by the BA does not change, the number of constraints increases with the number of users. As an example, to demonstrate the scalability of our approach, Fig. 12 shows the (x, y) plane with $J = 4$ devices being charged (represented by the black dots in the figure) and $U = 2$ users being avoided (represented by the green dots). In addition, $\delta_{\mathrm{EH}} = 350\ \mu$ W and $\delta_{\mathrm{U}} = 100\ \mu$W. As we observe, the proposed BA approach is still able to handle the optimization problem, meet all constraints for all devices.

Finally, an investigation about the size of the DMA has been carried out. Let us recall that the size of the DMA impacts the number of RF chains as $N_v = \left\lfloor \frac{L}{\lambda/2} \right\rfloor$. To illustrate this behavior in our 4 different scenarios, Fig. 13 plots the transmitted power required by the DMA as a function of L. As we observe, as L increases, the transmission power also increases for all scenarios. Let us also remark that the behavior of the variation of the parameter L was also studied in [12], with a similar behavior observed.

V. Conclusion

This paper proposed a BA-based optimization scheme to design the energy beamforming in NF-WPT systems powered by DMAs. Our results show the effectiveness of the proposed method in minimizing the transmission power of the DMA. In less complex scenarios where one or two users must be charged (respectively denoted by Scenarios 1 and 2), the proposed BA-based optimization converges fast, yielding lower transmission power compared to similar approaches from the literature. In addition, in scenarios where one device must be avoided, while one or two users are being charged (respectively denoted by Scenarios 3 and 4), which is not often considered in the literature, the proposed approach also has shown good performance, yielding similar transmission power requirements depending on the avoiding threshold. Complementarily, our results have also shown that stricter δ_{U} thresholds considerably increase the required transmission power.

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